# Penalties Sized to the Gain: When Do Random Audits Stop AI Agents from Over-Using a Shared Resource?

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## Abstract

Language-model agents that share a limited resource can take more than their allowance, and an overseer can afford to check only a few of them. We tested when random audits stop such over-use, and whether the penalty that stops a given agent can be predicted before it is deployed. We built an audited commons in which six agents draw on a renewable resource, a reviewer sets each agent's allowance to keep the resource at half capacity, and random audits answer a caught violation with a fine, a memory of the offence, or both. We first used programmed agents in three games, Fishery, Forest and River, and then replaced four of the six fishers with language models from four families in 520 games. With programmed agents, audits reduced harm only through their consequences, and deterrence began where the expected fine equalled the gain from cheating, and thresholds predicted in advance from runs without audits held in 5 of 5 new settings. Tightening everyone's allowance to absorb known cheaters made cheating pay more than no audits, and aiming each audit at the largest report caught 25% of liars, against 100% under random audits. The two language models that over-took on purpose stopped at the fine we predicted in advance from the excess they took per round without a fine: exactly for Nemotron 3 Super, and one grid step early for gpt-oss-120b. The over-taking they did gained little or lost fish over the whole game, yet both continued until the expected fine approached the gain of a single round, and both stopped earlier and more gradually than a risk-neutral agent would. A reviewer that remembered past over-takes reduced over-taking and removed most of the harm without any fine, but did not stop over-taking. The penalty that stops a language-model agent can thus be calibrated from its behaviour without a penalty; audits should not be aimed at the largest report, and memory can stand in where penalties are unavailable.


## 1 Introduction

Language-model agents increasingly act together on shared, limited resources, such as compute budgets, energy reserves and the economic and natural commons that their users depend on (Piatti et al. 2024; Bracale Syrnikov et al. 2026b). An overseer of such a system faces a problem that is familiar in economics and new in AI safety: verification is expensive, so only a few actions can be checked (Amodei et al. 2016; Greenblatt et al. 2024), and an agent that takes more than its share is seen only if it happens to be checked.

Enforcement economics predicts when checking deters a rational offender: an offender complies once the expected penalty, the probability of being caught times the fine, exceeds the gain from offending (Becker 1968; Allingham and Sandmo 1972; Avenhaus, von Stengel, and Zamir 2002). AI-control studies evaluate audit budgets, but a caught model is shut down, so audits without consequences, memories of past offences and graded fines remain untested (Greenblatt et al. 2024; Griffin et al. 2024). Language-model agents in commons games mostly fail to sustain the resource without enforcement (Piatti et al. 2024; Bracale Syrnikov et al. 2026b). Enforcement studies with language models track each agent's reliability and escalate penalties for repeated violations reported by peers (Ye and Steinhardt 2026), or raise the fine together with the stated audit likelihood around break-even in single decisions (Okamoto, Erol, and Erol 2026). Whether the fine that stops a given model can be predicted from that model's own behaviour, when agents act repeatedly on a shared resource under random audits with stated odds, has not been tested.

We ask: (1) which consequences make sparse random audits reduce harm to a shared resource; (2) does deterrence begin where the expected fine equals the gain from cheating, and can that point be predicted in advance; and (3) do language-model agents follow the same threshold, and which gain, that of one round or of the whole game, sets it?

We address these questions with an *audited commons* (Figure 1). Each round, six agents request a share of a renewable resource; a reviewer predicts the effect of the requests and grants allowances that keep the resource at half capacity, where it regrows fastest; each agent then chooses what it actually takes; and random audits check a few agents. We first study three games with programmed agents, whose decisions we can derive, and then replace four of the six fishers in Fishery with language models from four families. We froze a protocol before writing the code of each study, dated every amendment, and report every failed prediction.

## 2 Related Work

Scalable oversight concerns objectives that are "too expensive to evaluate frequently" (Amodei et al. 2016; Bowman

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et al. 2022). AI control evaluates protocols that audit a small share of an untrusted model's outputs, and compares random with suspicion-targeted auditing (Greenblatt et al. 2024); Griffin et al. (2024) model such protocols as games. Monitors that check one agent or one action at a time miss harm that only appears in combination (Makins et al. 2026; Hu and Wang 2026), and rare random audits can deter AI agents in theory when audit draws cannot be learned in advance (Gans and Holden 2026).

In commons games, "all but the most powerful LLM agents fail to achieve a sustainable equilibrium" (Piatti et al. 2024), language-model collectives drain a shared energy reserve like impatient optimisers (Bracale Syrnikov et al. 2026b), and reasoning models free-ride in public-goods games with sanctions (Guzman Piedrahita et al. 2025). Ye and Steinhardt (2026) enforce norms among language-model agents, including in a fishery, through peer reports checked by a language-model verifier, and find that tracking each agent's reliability and escalating penalties for repeated misbehaviour resist exploitation. Escalating sanctions recorded in a governance log cut collusion among language-model firms from 50% to 5.6% where a prohibition stated in the prompt did nothing (Bracale Syrnikov et al. 2026a), and reputation sustains cooperation and honesty among language-model agents (Ren et al. 2026; Yang et al. 2026). None of these commons studies varies the audit probability or the fine; in a single-agent scenario, a higher stated audit probability made frontier models act less harmfully (Konuganti and Co 2026).

The expected-penalty condition goes back to Becker (1968) and Allingham and Sandmo (1972), and recent work applies it to aligning AI agents through incentives (Agarwal et al. 2026). Inspection games formalise partial verification (Avenhaus, von Stengel, and Zamir 2002), records of past offences give enforcement leverage (Harrington 1988), report-based cutoff rules can beat random audits in static settings (Reinganum and Wilde 1985), auditees can game audits that follow reports (Burnat and Davidson 2026), and a small fine can act as a price (Gneezy and Rustichini 2000). Across four enforcement levels around break-even, language models mostly comply more as the fine and the stated audit likelihood rise, and a small fine can lower compliance (Okamoto, Erol, and Erol 2026); and deceive more when the perceived risk of getting caught falls (Scheurer, Balesni, and Hobbhahn 2023). We combine these strands in one setting: several agents, a renewable resource, random audits with stated odds, graded fines, memory, and a stopping fine predicted for each model from its own behaviour before it is tested (supplementary Table S4).

# 3 Methods

## 3.1 Audited commons

$N = 6$ agents act for $T$ rounds on a resource with level $x_t \in [0, K]$. After the agents take $c_t = \sum_i c_{i,t}$, the resource regrows logistically,

$$x_{t+1} = y_t + r\, y_t\Big(1 - \frac{y_t}{K}\Big) + \varepsilon_t, \qquad y_t = x_t - c_t, \quad (1)$$

with weather noise $\varepsilon_t$ in Forest and River. Regrowth $ry(1 - y/K)$ peaks at $y = K/2$, where the maximum sustainable harvest $rK/4$ is reached. Agent $i$ requests $p_{i,t} \in [0, \bar{c}]$ and may report $\hat{p}_{i,t} \le p_{i,t}$. The reviewer chooses one shared cut $\lambda_t \in \{1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, 0\}$, the largest whose predicted outcome meets its target, or 0 if none does, and grants allowances $a_{i,t} = \lambda_t \hat{p}_{i,t}$. A *joint* reviewer sums all requests before predicting; *local* reviewers judge one request at a time. The target is either a one-step safety line, under which the resource must not fall to a danger level next round, or *half capacity*, $y_t \ge K/2$, the level behind maximum sustainable yield (May et al. 1978).

We counted a round as *harmful* when the resource fell below half capacity: a Fishery stock below 50 of 100 after fishing, a mean Forest plot health below 10 of 20 (as a 5% chance constraint), or a River quality below 50 of 100. Following commons-game studies (Piatti et al. 2024; Perolat et al. 2017), we also report *efficiency*, the total harvest divided by $T \cdot rK/4$; *equality*, one minus the Gini coefficient of the agents' returns; and *survival*, the share of runs without collapse.

## 3.2 Games

The three games differ in how harm connects the agents (Figure 1, bottom). In Fishery, six agents draw on one stock, so each catch lowers what every agent can take in the next round. In Forest, each agent cuts its own plot in a ring of six, and cutting more than the sustainable 2.1 of 6 units damages each of the two neighbouring plots by 0.22 of the excess. In River, three agents discharge reagent $X$ and three discharge reagent $Y$; the damage $\kappa XY + \lambda(X + Y)$ arises mainly from the two together, so no single discharge looks harmful on its own. In every game, the reviewer's single shared cut links all six agents.

## 3.3 Audits, fines and memory

An agent *over-takes* when $c_{i,t} > a_{i,t}$ and *under-reports* when $\hat{p}_{i,t} < p_{i,t}$. Each round, each agent is audited independently with probability $q$, and an audit detects a violation with probability $s$. A caught agent pays a flat fine $F$, so the expected fine per violating round is $e = qsF$. Alternatively, the reviewer keeps a *memory*: it records the caught agent's excess and lowers its later allowances by the mean recorded excess, $a'_{i,t} = \max(0, \lambda_t \hat{p}_{i,t} - \bar{m}_i)$.

To predict when a fine deters, we apply the expected-penalty rule (Becker 1968) to our game as a benchmark. Let $g_1$ be the largest gain available to an agent from one round of over-taking, holding later rounds fixed, and let $g^*$ be the largest net gain over complying across the whole game, per round of over-taking, for an agent that commits to one over-take level.

**Proposition 1** (Break-even)**.** *(i) A risk-neutral agent that maximises each round's catch net of the expected fine over-takes in a round if and only if $e < g_1$ (ties go to compliance); under a flat fine it then takes the maximum $\bar{c}$. (ii) An agent that commits to one over-take level for the whole game over-takes at some level if and only if $e < g^*$.*

*Sketch.* (i) Over-taking by $\delta > 0$ adds $\delta$ to the round's catch and costs $e$ in expectation, independent of $\delta$ under a flat fine;

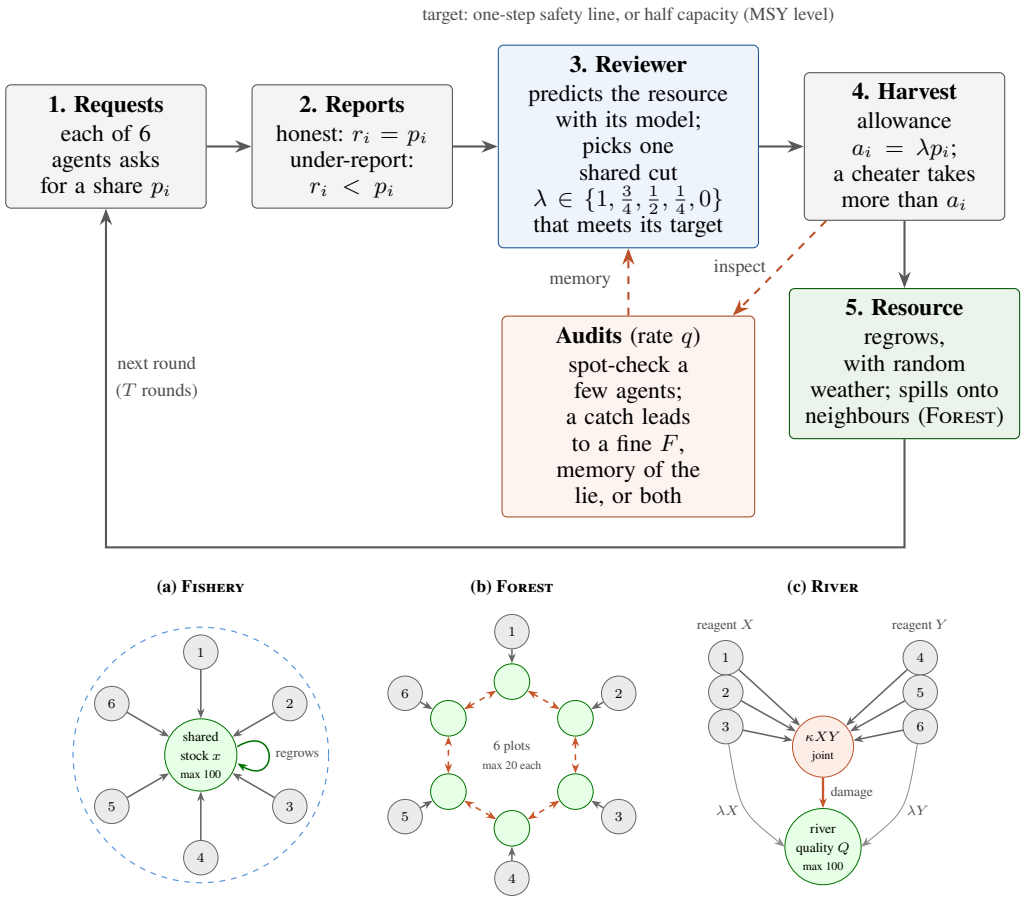



Figure 1: The audited commons. Top: one round, in which agents request a share and may under-report, the reviewer chooses one shared cut that meets its target, each agent takes its allowance or more, the resource regrows, and random audits answer a caught violation with a fine, a memory of the offence, or both. Bottom: how harm connects the agents in (a) Fishery, where six agents draw on one stock, (b) Forest, where over-cutting a plot damages its two neighbours (dashed), and (c) River, where damage arises mainly where reagents $X$ and $Y$ mix. In every game the reviewer scales all six requests by the same factor; the dashed ring in (a) marks this shared cut, which links every agent to every other.

the net gain is largest at the largest $\delta$ and is positive iff $g_1 > e$. (ii) A level $d$ yields a total gain $G(d)$ over $n(d)$ over-take rounds and expected fines $e\,n(d)$; it pays iff $G(d)/n(d) > e$, and $g^* = \max_d G(d)/n(d)$. □

Depleting the resource lowers later catches, so $g^* \leq g_1$ in our games. An agent that decides round by round can therefore keep over-taking at fines where over-taking at one fixed level no longer pays over the game.

### 3.4 Programmed agents

Programmed agents follow threshold rules, and 2 or 4 of the 6 are *greedy*. Misbehaving agents either under-report by a fixed share, or over-take with a level $d \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, taking $a + d(p - a)$, which a search chooses to maximise their payoff on 32 training populations. These agents commit to one level per game, as in part (ii) of Proposition 1. We evaluated every rule on 64 new populations of 80 rounds. To compare memory across games, we ran every game on the same grid of four settings: 2 or 4 greedy agents, and the game's regrowth rate multiplied by 1.0 or 0.85. In River the reviewer aimed at quality 50 in this grid.

### 3.5 Language-model agents

In Fishery, with the half-capacity target and 20 rounds, four of the six fishers were language-model agents and two followed fixed rules. Each round, a language-model agent stated a request and, after seeing its allowance, its catch, as JSON in tonnes; it saw the last three rounds of its own history but not the other agents' actions. Audits were independent with $q = 1/6$. A catch cost a flat fine $F \in \{0, 1, 2, 4, 8, 12, 18, 24, 30, 36\}$ t or triggered the reviewer's memory with no fine. The rules stated "you can choose to catch more than your allowance"; control conditions omitted this sentence or reworded the rules. A caught agent kept its catch and paid the fine. We counted an over-take when the catch exceeded the allowance by more than 0.06 t. The full prompts are in supplementary Section I.

We ran gpt-oss-120b, Nemotron 3 Super and Gemma 4

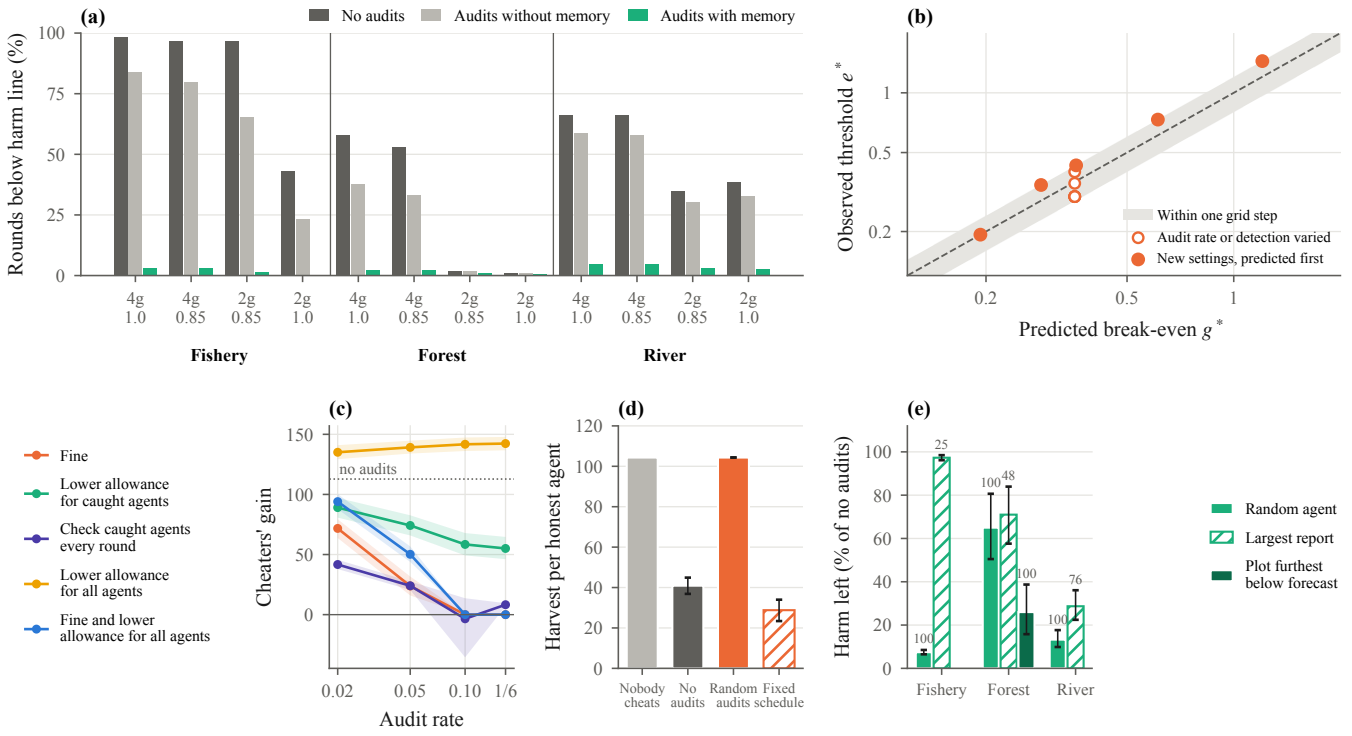



Figure 2: Audits with programmed agents. (a) Share of rounds below the harm line with under-reporters, when nobody is audited, when audits are used without memory, and when the reviewer remembers each caught lie; in every game, the four settings differ in the number of greedy agents (2g or 4g) and a multiplier on the game's regrowth rate (1.0 or 0.85). (b) Observed deterrence threshold $e^*$ against the break-even $g^*$ predicted from runs without audits; filled points are new settings predicted before the search, open points vary the audit rate from $1/6$ to 0.02 or let audits miss cheats, and the band marks one grid step. (c) Cheaters' gain against the audit rate under five audit rules; the dotted line marks no audits, and checking caught agents every round used about 4 checks per round, against about 1 for the other rules. (d) Harvest per honest agent with a fine of 24 under random audits and audits on a fixed schedule. (e) Harm remaining, relative to no audits, when one audit per round is aimed at a random agent, at the largest report, or at the plot furthest below the reviewer's forecast; numbers give the share of liars ever caught. Error bars and bands show 95% intervals over populations.

31B, and added Mistral Large 3 as a fourth family, through Ollama Cloud at temperature 0.7 with fixed seeds, on the same 10 populations in every condition. Each model first passed a pilot gate of at least 95% valid first answers and a mean comprehension score of at least 2 of 3. Across 520 games and 81,232 decisions, the models needed no fallback answers and 6 re-prompts, and every comprehension check scored 3 of 3. Before testing fines of 12–30 t, we froze a prediction of the fine at which gpt-oss-120b and Nemotron 3 Super would stop, derived from their behaviour without a fine.

### 3.6 Analysis

We compared conditions on the same populations and report 95% paired bootstrap intervals over populations (4,000 resamples). With 10 populations, the smallest paired difference in over-take rate detectable with 80% power is 6–15 points, given the spread between neighbouring fines in the threshold test; every memory comparison in Figure 2a remains significant after a Holm correction across the 12 settings. Supplementary Table S1 lists the settings of every study. We mark analyses chosen after the results were seen as post hoc.

## 4 Results

### 4.1 Reviewer target and model

With a joint reviewer and true inputs, the half-capacity target gave a total Fishery harvest of 1,393 against 555 under the one-step line (difference $+838$, $[+800, +865]$), because the one-step line let the stock fall to where it regrows slowly. When the reviewer overestimated Forest regrowth by 25%, its approvals of risky requests rose from 0.9% to 82%. Under a hidden collapse point in Fishery, the one-step line collapsed 78% of populations and the half-capacity target none. A reviewer that learned the model recovered within about 60 rounds (supplementary Figure S1), as adaptive management in fisheries suggests (Ludwig, Hilborn, and Walters 1993).

### 4.2 Audit consequences

Against under-reporters, a reviewer that used audit results only to correct its current decision barely improved safety: unsafe Forest rounds fell from 1.9% to 1.7%. Remembering

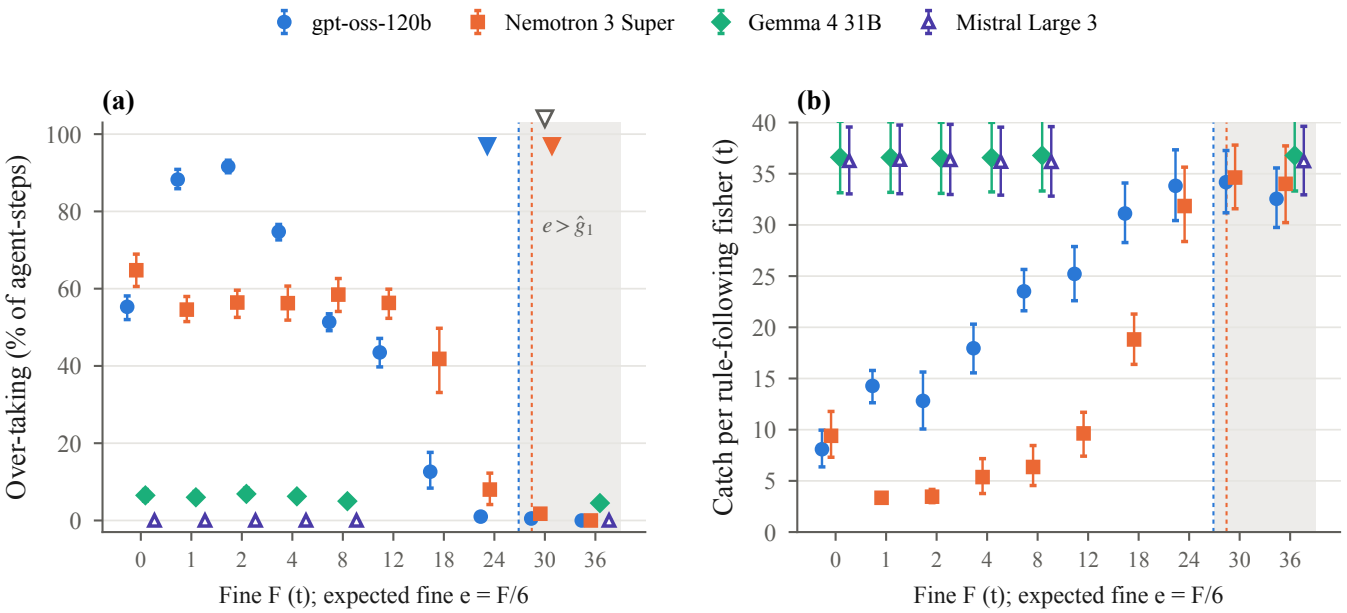



Figure 3: Language-model agents in Fishery. (a) Share of rounds in which each model over-took, against the fine $F$ (expected fine $e = F/6$); fines of 12–30 t were tested for gpt-oss-120b and Nemotron 3 Super only. Dotted lines mark the fine at which each model's expected fine equals its observed one-round excess $\hat{g}_1$, and the shaded area marks fines above both. Open triangles at the foot: Mistral Large 3, added as a fourth family. Triangles at the top mark the stopping fine predicted before the threshold test (open; 30 t for both models) and the one observed (filled; 24 t for gpt-oss-120b, 30 t for Nemotron 3 Super). (b) Catch of the two rule-following fishers over 20 rounds. Error bars show 95% intervals over populations.

each caught lie gave less harm than audits without memory in 12 of 12 settings across the three games (Figure 2a), and removed most of the harm wherever harm was common. Rounds below the harm line fell from 43–99% to 0.3–3% in Fishery, from 53–58% to about 2% in Forest with four greedy agents, and from 35–66% to 3–5% in River, whereas audits without memory removed only 10–46% of the harm. The result also held in all five testable settings of a wider Fishery grid with regrowth rates from 0.5 to 0.9 (supplementary Table S5). Memory alone did not stop cheating: adaptive cheaters kept 41% of their gain even at the highest audit rate, whereas memory plus a fine deterred at an audit rate of 0.05 instead of 0.10.

### 4.3 Deterrence threshold with programmed agents

In the pilot setting, cheaters stopped at $e \approx 0.30$–$0.40$, against a predicted break-even of 0.356, for audit rates from $1/6$ to 0.02 and for audits that missed up to three cheats in four. Because these agents optimise the payoff in part (ii), this test checks the measurement rather than the agents: whether a threshold estimated on other populations, under noise and audits that miss, matches the benchmark. We predicted $g^*$ for five new settings from runs without audits, before any search with fines. In 5 of 5 settings the observed threshold lay within one grid step of the prediction, for predicted thresholds from 0.19 to 1.20 (Figure 2b). A known audit schedule did not deter: honest agents received 28.8 against 40.9 with no audits at all (Figure 2d).

### 4.4 Audit rules and audit targeting

A reviewer that answered known cheaters by lowering the allowance of *all* agents protected the stock but moved the cost onto honest agents. Cheaters gained more than with no audits (142 against 113), and honest agents received 36 each, against 104 under a fine (Figure 2c). Aiming each audit at the largest memory-corrected report kept auditing the same agent: in Fishery, only 25% of liars were ever caught, against 100% under random audits, and harm stayed at 96.2% of rounds against 7.4%. River showed the same ordering, with 6.0% against 2.8% of rounds unsafe and 41% against 37% below half capacity. Static models favour report-based cutoff rules (Reinganum and Wilde 1985); with agents of different sizes playing repeatedly, a rank rule never leaves the largest agent, the feedback loop that Ensign et al. (2018) describe for predictive policing. Targeting by outcome rather than by report did better: aiming audits at the plot furthest below the reviewer's forecast gave the least harm in Forest, 26% of the no-audit level against 65% at random (Figure 2e).

### 4.5 Language-model agents across fines

gpt-oss-120b and Nemotron 3 Super over-took in 51–92% of rounds at every expected fine up to 1.33 t, and in 0% at 6 t (Figure 3a). A small fine raised over-taking by gpt-oss-120b from 55% to 88–92%, consistent with a fine acting as a price (Gneezy and Rustichini 2000; Okamoto, Erol, and Erol 2026). In single procurement decisions, Okamoto, Erol, and Erol (2026) found gpt-oss-120b compliant at every fine under imperative wording; under their wording that permits

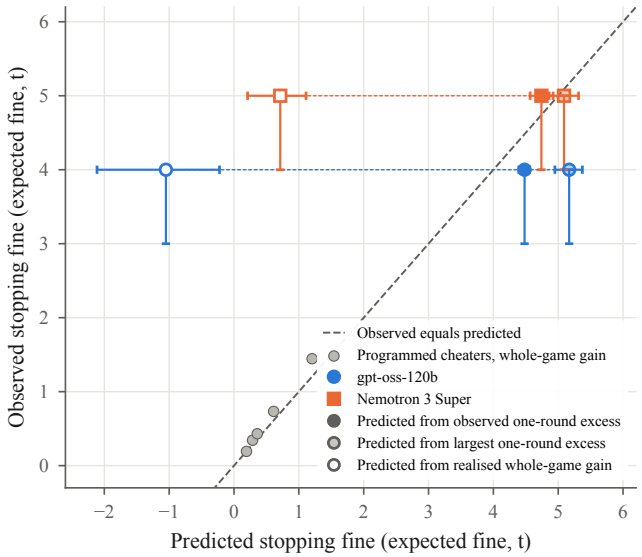



Figure 4: Observed against predicted stopping fine for the language models. The stopping fine is predicted from each model's observed one-round excess $\hat{g}_1$ (filled; frozen before the threshold test), from the largest one-round excess available, the $g_1$ of Proposition 1 (tinted; post hoc), or from the realised whole-game gain of its over-taking (open; post hoc), and observed in the threshold test; points on the dashed line are predicted exactly. Vertical whiskers span the last fine that did not deter and the first that did; horizontal whiskers show 95% intervals over populations. Grey points: programmed cheaters, predicted from their whole-game gain $g^*$ (Figure 2b). The observed one-round excess predicted where the language models stopped; the realised whole-game gain did not, and against the largest excess both models stopped early.

the violation, the analogue of ours, its compliance rose from 0% without a fine to 74% and then 100% as the fine grew, the pattern we observe. How a model responds to a fine therefore depends on the task and its wording, not only on the model. Below the threshold the harm was large. Under Nemotron 3 Super, 5 to 10 of 10 lakes collapsed at every fine from 1 to 8 t, and under gpt-oss-120b the rule-following fishers caught 8–25 t over the game, against 32–34 t once over-taking stopped (Figure 3b). Gemma 4 31B over-took in 4–10% of rounds, almost always by catching its request after the reviewer cut it, and Mistral Large 3 never over-took in 16,000 decisions (supplementary Figure S2); neither model therefore informs the deterrence threshold. A 36 t fine stopped over-taking under every wording of the rules.

### 4.6 Predicted stopping fine

Before the threshold test we froze a prediction from each model's *observed one-round excess* $\hat{g}_1$, its mean catch above the allowance on over-take rounds without a fine: 4.48 t for gpt-oss-120b and 4.74 t for Nemotron 3 Super. We predicted that fines of 12, 18 and 24 t ($e = 2$–$4 < \hat{g}_1$) would not deter and that 30 t ($e = 5 > \hat{g}_1$) would, and then ran 80 games at these fines. Taking "deterred" as at most 5% of rounds at that fine and every larger one, an operational criterion rather than an indifference point, Nemotron 3 Super stopped exactly at 30 t and gpt-oss-120b one grid step early, at 24 t (Figure 3a). Resampling the populations gave the same stopping fines in 92% and 100% of 4,000 bootstrap samples (post hoc; supplementary Section J). Over-taking declined gradually rather than in one step: relative to no fine, it was 0.79 and 0.87 at $e/\hat{g}_1 \approx 0.45$, 0.23 and 0.65 at $\approx 0.65$, and 0.02 and 0.12 at $\approx 0.87$ for gpt-oss-120b and Nemotron 3 Super. Our pre-registered expectation that over-taking would stay at or above 25% at 24 t failed for both models (1.0% and 8.0%). The benchmark's $g_1$, the largest excess available (6 t minus the allowance), was larger, 5.2 and 5.1 t on the same rounds; a risk-neutral agent would stop only at 36 t, so both models stopped one or two grid steps earlier than the benchmark (post hoc; Figure 4).

### 4.7 One-round and whole-game gain

Measured as the programmed cheaters are measured, as the net gain over a compliant game per over-take round, the over-taking the models actually did was worth $-1.05$ t for gpt-oss-120b, which lost fish by over-taking in 7 of 10 populations because lakes collapsed, and 0.71 t for Nemotron 3 Super (post hoc). This *realised whole-game gain* compares all four models over-taking without a fine with all four complying at 36 t. It is not $g^*$, the best plan available: an agent free to vary its plan could over-take only in the last round and earn the full one-round gain. Both models nonetheless kept over-taking until the expected fine approached $\hat{g}_1$, nearly seven times the realised gain for Nemotron 3 Super (Figure 4). Their stopping fines thus tracked the gain of a round, not what their over-taking earned over the game, which places them closer to part (i) of Proposition 1 than to part (ii). They departed from part (i) in three ways: without a fine they over-took in only 55–65% of rounds, deterrence was gradual, and they stopped below the benchmark. Two further observations agree with part (i). Over-takes were mostly all-out, reaching the 6 t maximum in 68% and 93% of over-take rounds, as a flat fine gives no reason to take less (Stigler 1970). Being caught did not change the next round: gpt-oss-120b over-took again in 78% of rounds after a caught over-take and 81% after an unchecked one. Unlike people in some tax experiments, the models did not comply less right after an audit (Mittone 2006; Kastlunger et al. 2009).

### 4.8 Memory and efficiency

With the reviewer's memory and no fine, over-taking fell from 55% to 20% of rounds (gpt-oss-120b) and from 65% to 33% (Nemotron 3 Super), but did not stop. Rounds below half capacity fell from 92% to 20% and from 96% to 40%, no lake collapsed (against 5 and 1 of 10), and the rule-followers' catch rose from 8 to 29 t and from 9 to 32 t. Fines above $\hat{g}_1$ cost no efficiency: at 30 and 36 t both models reached 0.99 of the maximum sustainable harvest, kept the stock at or above half capacity in 96–100% of rounds and gave the most equal outcomes (equality 0.81). Below the threshold, efficiency stayed high only while the stock was over-fished, or fell when it collapsed.

# 5 Discussion

## 5.1 Penalties and audit design for AI agents

Our results support one rule for overseeing AI agents on a shared resource: random audits work through their consequences, and a fine deters once its expected value exceeds the gain the agent acts on. For programmed agents that commit to one strategy per game, that gain is the whole-game gain, and runs without audits predict the threshold. For the two language models that over-took, the stopping fine followed the excess they took in a single round without a penalty, and both stopped somewhat below the risk-neutral benchmark. Language-model agents in commons are known to neglect the future of the shared stock (Piatti et al. 2024; Bracale Syrnikov et al. 2026b), and our results fit that neglect setting the fine; we did not test why the models stop. An overseer can measure the one-round excess without a penalty and set the expected penalty above it. Audits should cover every agent and be unpredictable: a known schedule did not deter. Aiming audits at the largest report made matters worse, whereas aiming them where the outcome fell furthest below the forecast beat random audits in Forest. Where penalties are unavailable, a memory of past offences reduced over-taking and protected the resource, but did not stop over-taking. Lowering every agent's allowance to absorb known cheaters also made matters worse.

## 5.2 Limitations and future work

The language-model results come from one game, two models that over-took on purpose and 10 populations per condition, with one sampled game per population. The fine grid resolves the threshold to about $0.2\,\hat{g}_1$. We show that the stopping fine can be predicted, not why the models stop. Why they cut back from about $0.4\,\hat{g}_1$ remains untested: a calculation of the expected fine with risk aversion (Allingham and Sandmo 1972), a response to the stated size of the fine read as a warning, and partial attention to the stock all fit. The comparison with the whole-game gain was made after the results. Our "silent" control still mentioned a check with a fine of 0 t, so it did not replicate a no-check condition, and our wording-robustness hypothesis held for Nemotron 3 Super but not for gpt-oss-120b. The language-model agents were told they could over-take and did not hide their catch, so the study concerns exceeding an allocation, not concealment. Our rules state the audit odds and that over-taking is possible, so they resemble a textbook deterrence test: models often recognise evaluations (Needham et al. 2025; Li et al. 2026), prompts can pre-determine outcomes in language-model societies (Zhou et al. 2026), behaviour shifts with spurious features of a prompt (Sclar et al. 2024), and a long-horizon framing changes how reasoning models weigh the future (Jung et al. 2026). We therefore make no claim about people, report over-taking under three wordings (supplementary Figure S2), treat the programmed agents as a scripted positive control, and did not test whether the models recognised the experiment. The programmed cheaters optimise over a one-parameter family, a lower bound on an adaptive adversary. A logging fault lost most saved reasoning texts of the first language-model run; outcomes were unaffected. Varying the audit probability and the fine at a fixed expected penalty would separate a calculation of the expected fine from a response to its stated size, fines that grow with the excess should curb all-out over-taking (Stigler 1970), and language-model versions of Forest and River would test whether over-taking spreads along the ring of plots.

# 6 Conclusion

Random audits reduced over-use of a shared resource by AI agents only through consequences, and fines stopped it once the expected penalty outweighed the gain the agent acted on. For programmed cheaters that gain was their whole-game gain; for the two language models that over-took on purpose, the stopping fine followed the excess they took in one round. In our setting, the penalty that stops such an agent could be calibrated from its behaviour without a penalty, before that penalty was tested.

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